

11(a)	energy (of photon) required to remove electron	M1
	from a surface or reference to <u>minimum</u> energy or reference to zero <u>kinetic</u> energy	A1
11(b)(i)	1. photon energy = hc / λ	C1
	$= (6.63 \times 10^{-34} \times 3.00 \times 10^8) / (280 \times 10^{-9})$ $= 7.1 \times 10^{-19} \text{ J}$	A1
	2. electron energy = $(7.1 - 5.5) \times 10^{-19} \text{ J}$	C1
	$\frac{1}{2} \times 9.11 \times 10^{-31} \times v^2 = (7.1 - 5.5) \times 10^{-19}$	C1
	$v = 5.9 \times 10^5 \text{ m s}^{-1}$	A1
11(b)(ii)	energy is required to bring electron to the surface	B1
11(c)	no change	B4
	increases	